

NAGO, Japan

WMO: 479400

Lat: 26.5942N

Lon: 127.9664E

Elev: 23

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.1	52.9	36.4	31.5	54.9	38.5	34.3	56.3	21.4	59.8	20.1	60.1	7.3	350	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	90.4	80.0	89.5	79.7	88.5	79.5	81.9	87.0	81.3	86.5	80.8	86.1	9.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.6	158.9	85.0	79.9	155.2	84.7	79.2	151.8	84.5	45.7	87.3	45.1	86.6	44.5	86.3	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.3	19.8	17.4	DB	47.4	92.4	2.2	1.2	45.9	93.3	44.6	94.0	43.4	94.6	41.8	95.5
			WB	43.1	82.8	2.2	0.9	41.5	83.4	40.2	84.0	38.9	84.5	37.3	85.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.4	61.4	62.8	65.1	69.7	75.2	80.6	84.6	84.5	82.5	77.5	71.6	64.8
	DBStd	9.04	4.32	4.90	4.78	4.05	3.58	3.46	1.66	1.72	2.10	3.43	3.91	4.38
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	350	127	93	61	8	0	0	0	0	0	0	3	58
	CDD50	8551	355	359	467	591	782	919	1072	1070	976	853	647	460
	CDD65	3425	17	32	63	149	317	469	607	605	526	388	199	53
	CDH74	31194	3	14	47	337	1750	4612	7560	7493	5847	2845	632	54
	CDH80	10167	0	0	0	6	175	1285	3152	3102	1927	493	27	0
(14) Wind	WSAvg	8.4	8.4	8.6	8.5	7.8	7.1	8.7	8.7	8.2	8.2	9.4	8.6	8.7
(15) Precipitation	PrecAvg	80.1	3.9	4.4	5.4	6.5	8.9	10.4	6.8	9.3	9.1	6.6	4.7	4.2
	PrecMax	133.3	9.6	17.0	13.9	13.9	21.1	26.4	19.6	27.5	25.4	21.0	10.9	12.1
	PrecMin	52.5	0.7	1.0	1.2	1.3	0.8	2.5	0.4	1.9	0.4	1.5	0.4	1.1
	PrecStd	18.3	2.0	3.3	2.8	3.6	4.8	5.6	5.3	6.1	6.3	4.4	2.8	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.5	75.6	77.3	81.0	85.4	89.1	91.5	92.2	90.6	88.2	82.6	78.0
		MCWB	68.1	69.6	71.1	74.6	78.5	80.7	80.1	80.5	79.6	78.1	74.6	72.1
	2%	DB	72.3	73.8	75.3	79.0	83.5	87.9	90.3	90.7	89.1	86.1	80.7	75.5
		MCWB	66.0	68.2	69.2	73.3	76.9	80.0	79.9	80.2	79.0	77.4	73.2	68.7
	5%	DB	70.5	72.3	73.8	77.4	82.0	86.8	89.3	89.6	88.0	84.3	79.1	73.6
		MCWB	64.1	66.8	67.9	71.9	75.6	79.6	79.7	79.9	78.6	76.2	72.3	67.0
	10%	DB	68.7	70.7	72.3	76.0	80.5	85.6	88.4	88.5	86.8	82.7	77.6	72.0
		MCWB	62.3	65.3	66.7	70.5	74.7	79.2	79.6	79.6	78.1	75.1	71.3	65.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.0	71.0	72.5	76.5	80.0	82.2	82.5	82.8	81.8	80.0	76.5	73.7
		MCDB	73.3	74.6	75.8	79.4	83.3	87.1	87.5	88.1	87.5	84.6	80.1	76.6
	2%	WB	67.3	69.7	70.9	74.7	78.2	81.2	81.7	81.9	80.9	79.0	75.2	70.8
		MCDB	70.9	73.0	74.3	77.9	81.8	85.7	87.0	87.2	86.2	83.8	79.2	74.0
	5%	WB	65.3	68.0	69.3	73.3	77.0	80.5	81.1	81.4	80.2	77.9	73.9	68.6
		MCDB	69.2	71.3	72.7	76.6	80.7	84.9	86.5	86.8	85.4	82.8	78.1	72.4
	10%	WB	63.5	66.3	67.7	71.9	76.1	79.9	80.6	80.7	79.6	76.8	72.4	66.6
		MCDB	67.9	70.2	71.4	75.2	79.8	84.2	86.1	86.4	84.7	81.7	76.9	71.1
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	8.9	9.4	9.2	8.7	7.2	7.9	8.4	8.8	8.5	8.5	8.6
		MCDBR	10.8	10.1	10.0	9.4	9.2	7.6	9.0	9.7	9.7	9.5	9.6	10.4
	5% WB	MCWBR	7.4	7.0	7.0	5.8	4.8	3.2	3.2	3.5	3.8	4.2	5.3	6.9
		MCWBR	9.4	8.7	8.8	8.1	7.9	7.3	7.9	8.3	8.5	8.2	7.9	8.7
(40) Clear-Sky Solar Irradiance	taub		0.461	0.492	0.564	0.572	0.518	0.475	0.474	0.491	0.488	0.485	0.468	0.459
		taud	2.116	2.052	1.887	1.926	2.142	2.329	2.330	2.269	2.246	2.194	2.194	2.129
	Ebn at Noon		244	247	237	239	252	261	261	256	254	247	241	238
		Edh at Noon	44	50	62	62	49	41	41	43	43	43	40	42
(44) All-Sky Solar Radiation	RadAvg		758	948	1187	1393	1542	1693	2072	1890	1652	1287	933	760
	RadStd		91	144	115	124	201	153	165	123	117	108	87	73

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	+1.35	+1.82	N/A	N/A	N/A	N/A
Regional (6 neighbors)		N/A	N/A	N/A	N/A	+0.74	+0.95	N/A	N/A	N/A	N/A

Nomenclature: See separate page